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EE 446 — TinyML — Homework 1 Discussion

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1(e). Can you further reduce the model size beyond parts (b), (c), and (d)?

Yes. I combined all three techniques: I took the small student model from part (d), applied pruning to it during training, and then quantized it to int8.

Results from each part:

- Baseline (a): 14.08 KB, 98.15% accuracy
- Best quantized (b, int8): 8.00 KB, 98.15% accuracy
- Pruned (c): 8.62 KB, 92.59% accuracy
- KD student (d): 6.12 KB, 90.74% accuracy
- **Combined (KD + pruning + int8): 4.84 KB, 96.30% accuracy.**

The combined model is smaller than any single technique and is also more accurate than the pruned-only or KD-only models.

Why it works: each technique shrinks the model in a different way, so they stack. KD makes the network smaller (fewer layers and neurons). Pruning sets most of the remaining weights to zero. Int8 quantization stores each weight in 1 byte instead of 4. The teacher's soft labels also give the student extra information during training, which is why accuracy goes up instead of down.

Limit: below about 5 KB, most of the TFLite file is metadata (operator definitions, tensor shapes, quantization tables), not weights. To go smaller I would need a different model type or the EON Compiler.

2.1. What types of Learning Blocks are available under the free-tier Edge Impulse account?

The free tier has two main categories: supervised learning and unsupervised learning. There is no self-supervised learning block.

- **Supervised — classification:** Classification (NN). Example: gesture recognition from accelerometer data.
- **Supervised — regression:** Regression. Example: predict water flow rate in L/min from a flow sensor.
- **Supervised — object detection:** MobileNetV2 SSD FPN and FOMO. Example: count objects with an ESP32-CAM.

- **Supervised — transfer learning (images):** MobileNetV1/V2, EfficientNet. Example: classify a plant vs. a lamp.
- **Supervised — transfer learning (audio):** Keyword Spotting. Example: a “yes/no” voice command model.
- **Supervised — classical ML:** scikit-learn-style models. Example: random forest on tabular sensor features.
- **Unsupervised — anomaly detection:** K-means and GMM. Example: detect an unusual vibration on a motor.
- **Unsupervised — visual anomaly:** FOMO-AD. Example: spot defective parts on a conveyor belt.

2.2. List all the types of layers that can be used under the “Classifier” section in the free-tier Edge Impulse. Briefly explain the purpose of each layer.

- **Input Layer:** sets the shape of the data going in.
- **Dense (fully connected):** every neuron connects to every input. The main layer for general learning.
- **Conv1D:** 1-D convolution for time-series data like audio or accelerometer signals.
- **Conv2D:** 2-D convolution for images and spectrograms.
- **DepthwiseConv1D / DepthwiseConv2D:** convolution done one channel at a time. Uses far fewer parameters.
- **MaxPooling1D / MaxPooling2D:** shrinks the feature map by keeping the largest value in each window.
- **AveragePooling2D:** shrinks the feature map by averaging values in each window.
- **Flatten:** turns a multi-dimensional feature map into a 1-D vector so a Dense layer can use it.
- **Reshape:** changes the shape of the data without changing the values.
- **Dropout:** randomly turns off some neurons during training to prevent overfitting.
- **Batch Normalization:** rescales activations to help the model train faster and more stably.
- **TimeDistributed:** applies the same layer to every time step of a sequence.
- **ReLU:** activation function, outputs $\max(0, x)$. Standard between hidden layers.
- **Softmax:** activation function that turns the final outputs into class probabilities.

2.3. What types of model compression options are available under the “Deployment” section in the free-tier Edge Impulse?

The free tier offers two choices that combine to give four options:

Model version:

- **Unoptimized (float32):** weights stay in 32-bit floating point. Most accurate, largest size.
- **Quantized (int8):** weights and activations are converted to 8-bit integers. About 4× smaller, faster on MCUs.

Compiler:

- **TensorFlow Lite for Microcontrollers (TFLM / LiteRT):** the standard interpreter-based runtime.
- **EON Compiler:** turns the model into C++ source code with no interpreter. Saves about 25–55% RAM and 25–35% flash.

(The EON Compiler — RAM optimized option is enterprise-only and not in the free tier.)

2.4. What input modalities are supported for synthetic data generation in the free-tier version of Edge Impulse? List the methods. Why is synthetic data important?

Edge Impulse supports three input modalities for synthetic data:

- **Images:** DALL-E 3 (OpenAI) for text-to-image generation, and the syntheticAIdata vision-dataset block.
- **Audio:** ElevenLabs for sound effects (e.g., glass breaking, dog barking), and Google Cloud Text-to-Speech for spoken keywords.
- **Time series:** PyBullet for physics-simulated IMU/accelerometer data, and MATLAB for user-scripted signals.

Why synthetic data matters:

- **Saves time and money:** collecting thousands of real samples can take weeks; a generator does it in minutes.
- **Handles rare events:** failures and anomalies are rare in real life, but you can generate as many as you need.
- **Protects privacy:** synthetic voices and faces don't involve real people, so no GDPR or HIPAA issues.
- **Covers edge cases:** you can ask for unusual lighting, accents, or sensor conditions on demand.
- **Grows small datasets:** mix a small real dataset with a larger synthetic one to bootstrap the model.

The main downside is the *domain gap*: synthetic data may not perfectly match real-world conditions, so models should still be tested on real data.